

(Best Calendar Canvas (BCC))
(Team 3)
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Software Requirements Specification Document

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Changes

- Removed the manual cross out function specified in purpose
- Removed Google Chrome extension stretch goal 2.2
- Added adding and deleting assignments to goals 2.2

- Removed some attributes of teacher that are not necessary since the calendar will most likely not work with teachers access codes 3.2
- Removed login system implementation for chrome extension
- Removes username and password implementation for chrome extension
- 6.3.2 changed the nonfunctional requirement to fit the assignment weight instead of progress bar since it made more sense given the implementation
- 6.3.2 changed the nonfunctional requirement so that assignments now show as deleted after switching the page instead of within 1 second
- 6.3.3 changed software to system for the API calls requirement
- 6.3.4 deleted part in requirement where the user needs to login to canvas to use the software as now only an access token is needed
- 4.1 changed use case diagram to be more accurate
- Removed all mention of google chrome extension as this stretch goal was not achieved
- 6.3.2 changed the nonfunctional requirement for deleted assignments back as we were able to fix the issue causing the assignment not to be deleted until pages were changed

1 Purpose

This section describes the business background (or problem context).

Canvas was created to allow more efficiency in transferring information and announcements between teachers and students online where they do not need to be in the same room. Students need to use canvas to access course details and assignments. Professors assign work through canvas which includes discussions, assignments, and announcements. It is up to students to keep track of all their assignments and the progress made in each class. It is also up to students to keep track of the weight of each assignment.

2) why a new system is needed. You should describe the existing problems, issues, or deficiencies in the business where the use of the new system can bring business values (by addressing the problems, issues, or deficiencies).

Understanding that all students will have multiple teachers and receive multiple assignments from different classes, it can be hard to prioritize assignments especially those that may be worth a significant amount of the total grade. Additionally, it is important to not lose track of when assignments are due so the students can be more organized and efficient with their course work. Students may have a hard time visually understanding the impact that each assignment may have in a class from the simplistic calendar provided by Canvas. Canvas does not provide a method of checking the progress made in assignments separated by class. The new system can fix this by providing a color-coded calendar that displays the assignment weight and displays bars that track assignment completion separated by class.

2 Scope

Describe the scope of the software by clearly list the desired objectives. This sets up a scope for the new system to be developed. You may choose to address only a few of the problems, issues, or deficiencies identified in the business. In particular, you should

1) Give an appropriate name to the system (e.g., PSU Campus Map, Super Team Editor, etc.) and reference it by name in the rest of the document;

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2) Explain goals (what the software will do);

Goals:

- Show an assignment's weight and overall impact on grade using color
- Progress bars to track progress of assignment completion in each class
- Add and deleting assignments
- Show accurate calendar days with accurate assignment due dates

3) Describe the application of the software, including potential benefits.

When applied, the software will give students a new arrangement of functions and more opportunity to better organize their assignments. Potential benefits will include an increase in student activity on canvas and corresponding assignments as they will use canvas more to help keep track of their assignments and better visualize their work plan. Another benefit is with the added help with the calendar, students will most likely be able to complete assignments more efficiently and on time.

3 User characteristics

Identifying the potential users of the product. Describe general characteristics of the intended groups of users (stakeholders) of the product, especially focusing on characteristics that may influence usability, such as educational level, experience, disabilities, and technical expertise.

3.1 Key users

They are critical to the continued success of the product. Give greater importance to requirements generated by this category of user.

- *User role responsibilities: what to do with the product.*
- *Subject matter experience: Summarizes the users' knowledge of the business (domain).*

Rate as novice, journeyman, or master.

- *Technological experience: Describes the users' experience with relevant technology.*

Rate as novice, journeyman, or master.

- *Other user characteristics: Describe any characteristics of the users that have an effect on the requirements and eventual design of the product. For example:*

- o *Physical abilities/disabilities*
- o *Intellectual abilities/disabilities*
- o *Attitude toward technology*
- o *Education*
- o *Linguistic skills*
- o *Age group*
- o *Gender*

Key users of this application would be students who already use canvas or who are using canvas for the first time and are looking for ways to visually enhance their calendar as well as make their calendar more accurate and efficient.

- Students enrolled in a school or university that uses canvas as their main assignment outlet
- Students will be able to use BCC to easily determine which assignments to prioritize and which are completed
- Student is a journeyman of the domain
- Student should be able to use a computer and be able to login and access canvas
- Student is responsible for manually deleting completed assignments that do not already show up as completed on the calendar
- Student should be able to identify the difference in assignment weight based on visual representation
- Color Blindness would affect ability to view assignment prioritization feature
- Student may have already been enrolled in courses that uses canvas or may be using canvas for the first time
- Student should be able to read

3.2 Secondary users

A secondary user for this software may be professors or teachers that use canvas to assign homework and assignments to their students.

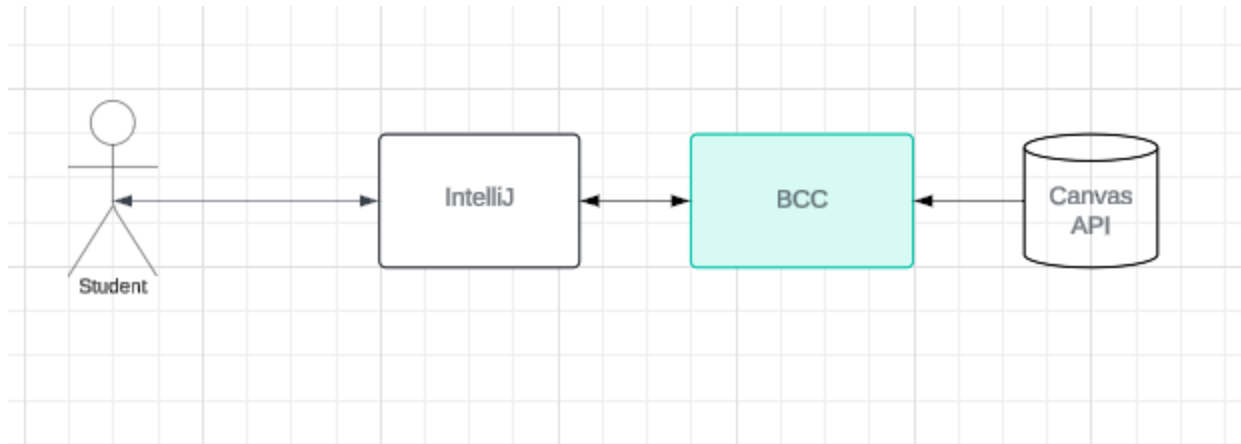
- Teachers knowledge of canvas may vary from novice to journeyman
- Teachers are responsible for providing accurate data when it comes to grading and marking assignments as completed
- Teachers are responsible for assignments having the correct due date and points
- Teachers should be able to use a computer or smart device to access canvas
- Teachers should be able to read and write in English

4 Product perspective

4.1 System Context

Define the system's relationship to users or other related systems. If the system is an element of a larger system, then identify the interfaces between the system covered by this SRS and the larger system. A block diagram showing the major elements of the larger system, interconnections, and external interfaces can be helpful. Below are two example block diagrams (with the system to be developed highlighted):

Blue = System to be developed



4.2 User interfaces

Specify the required characteristics of each interface between the software product and its users. Keep in mind that we are specifying requirements, so we here only care about the requirements on interfaces (this is NOT the place to put user interface designs by the software developers). For each requirement on an interface, it may include:

- *Required screen formats (text only or multi-media),*
 - *Required page or window layouts,*
 - *Required content of any reports, menus, error messages,*
 - *Required backup and recovery operations, or*
 - *Required user-initiated operations or function keys.*
-
- It is required that the screen format shall be multi-media
 - It is required that the content shown on calendar should be from canvas API
 - It is required that the page should show accurate calendar month and progress bars
 - It is required that system should allow users to delete completed assignments manually or add assignments
 - It is required that the system keeps record of completed assignments per course
 - It is required that the assignment color bar changes according to assignment weight

4.3 Software interfaces

- A required systems that need to be used for this software is the Canvas API
- A required system the customer needs as a prerequisite for this software is IntelliJ or another compatible Java IDE

4.4 Deployment requirements

- Customer should already have IntelliJ or another compatible Java IDE installed on their device
- Customer should have at least a download speed of at least 5 MB/s
- Customer should have at least 3 GB of RAM free for the extension to run properly

5 Assumptions and Dependencies

List each of the factors that affect the requirements stated in the SRS. These factors are not design constraints on the software but any changes to these factors can affect the requirements in the SRS.

For example, an assumption may be that a specific operating system will be available on the hardware designated for the software product. If, in fact, the operating system is not available, the SRS would then have to change accordingly.

- Canvas might crash which would make the application unable to access assignments
- The professor must continuously upload all assignments to Canvas with accurate point values and due dates
- If an assignment does not have a due date it will not be posted to calendar
- It is assumed user is a student
- It is assumed professors have accurate class schedules and assignment due dates
- It is assumed that professors will update assignments as completed once turned in by the student
- It is assumed that all users have access to the internet

6 Specific requirements

This is a specification, a designer should be able to read this spec and build the system without bothering the customer again.

- *Specify all of the software requirements to a level of detail sufficient to enable designers to design a software system to satisfy those requirements.*

- *Specify all of the software requirements to a level of detail sufficient to enable testers to test that the software system satisfies those requirements.*
- *Each requirement should be uniquely identifiable for traceability. Usually, they are numbered R.1, R.1.1, R.1.2.1 etc. so that each can be cross-referenced in other documents.*
- *Each requirement should also be testable. Avoid imprecise statements like, "The system shall be easy to use." What is easy to use?*

Use proper terminology:

- *A required, must have feature: The system shall...*
- *An optional, nice-to-have feature that may never make it to implementation: The system may...*

Avoid over-constraining your design. Do not require specific software packages, etc., unless the customer specifically requires them.

Avoid examples: Don't say things like, "The system shall accept configuration information such as name and address." The designer doesn't know if that is the only two data elements or if there are 200. List every piece of information that is required so the designers can build the right UI and data tables.

6.1 System Functional Requirements

Provide a list of the major functions that the software will perform. For example, an SRS for an accounting program may use this part to address customer account maintenance, customer statement, and invoice preparation without mentioning the vast amount of detail that each of those functions requires.

As mentioned in lecture, you should try the best to use the User Story Template to document system functional requirements.

- *As a student I want to each assignment to be color coded so that I can better manage my assignments and visualize the weight of each assignment*
- *As a student I want to be able to delete or add assignments so that I can maintain an accurate assignment to do list*
- *As a student I want to be able to see the progress made in each class so that I can better prioritize classes and the corresponding assignments*

6.2 Logical Database Requirements

Specify the logical requirements for any information that is to be placed into a database, including data entities and their relationships, and Integrity constraints.

If the customer provided you with data models, those can be presented here. ER diagrams (or static class diagrams) can be useful here to show entity relationships. Remember a diagram is worth a thousand words of confusing text.

- *Database would store added assignments as a stretch goal*
- *No database is required for this application*

6.3 Software System Attributes

*Specify non-functional requirements, that is, the required attributes of the software product.
For each requirements in this section, make sure it is testable*

6.3.1 Usability

Usability requirements for the software system include measurable effectiveness and satisfaction criteria in specific contexts of user interactions.

- The system shall provide accurate information of due dates per a given assignment from courses per the month and week
- The system shall provide accurate progress made per course
- The system shall show on the calendar completed assignments
- The system shall allow the user to be able to delete assignments from the calendar
- As a student, I would like the progress bar to fit nicely in the page and not look out of place.

6.3.2 Performance

Specify both the static and the dynamic numerical requirements placed on the software or on human interaction with the software as a whole.

- *Static numerical requirements (capacity) may include the following the number of simultaneous users to be supported; amount and type of information to be handled.*
- *Dynamic numerical requirements may include, for example, the numbers of transactions and tasks and the amount of data to be processed within certain time periods for both normal and peak workload conditions.*

The performance requirements should be stated in measurable terms. For example, 95 % of the transactions shall be processed in less than 1 second.

- The system shall integrate the assignments weight into the page at the same time of loading the page
- Assignments that are deleted shall not be shown on the calendar after 1 second

6.3.3 Reliability/Dependability

Specify the factors required to establish the required reliability/dependability of the software system at time of delivery.

- System must be able to make API calls
- Must provide accurate weights for class assignments
- The calendar application should be accessible for viewing and modifying all hours of the day in which canvas is also functional

6.3.4 Security

Specify the requirements to protect the software from accidental or malicious access, modification, or destruction. Specific requirements in this area could include the need to:

- 1) *Utilize certain cryptographic techniques;*
- 2) *Keep specific log or history data sets;*

- 3) *Authenticate system users;*
- 4) *Check data integrity for critical variables;*
- 5) *Assure data privacy.*

- The system will not collect personal information from students
- The system shall keep track of current and previous assignments both ongoing and completed for given course for as long as a user is enrolled in that course
- The system shall not authenticate users however the system shall only run once the user has provided a valid access token

6.3.5 Maintainability

Specify attributes of software that relate to the ease of maintenance of the software itself. These may include requirements for certain modularity, interfaces, or complexity limitation.

Requirements should not be placed here just because they are thought to be good design practices.

- We will set limitations on how much complexity each module of coding shall have, ensuring that each module is easier to understand, letting us have an easier time fixing or maintaining the code in the future. An example of this would be creating a separate module for finding due dates instead of building it into the assignment weight module, so that when Canvas is inevitably updated in the future it can be easily changed.
- All code should be well described in terms of comments or text to describe the module codes so altering them is easy. An example of this would be commenting the formula for assignment weight calculation before the line of code that implements it
- Each module should be testable to some degree to allow quick identifying of problems or errors. An example of this would be allowing the assignment bar module to be run separately of the other modules so that testing it individually will be simple to do.